



# AIRLINE PASSENGER **SATISFACTION** ANALYSIS

Empowering the Aviation Industry with Production-Ready Machine Learning & Interactive Analytics

DATA PIPELINE • PREDICTIVE MODELING • BUSINESS INSIGHTS

# PROJECT INFRASTRUCTURE

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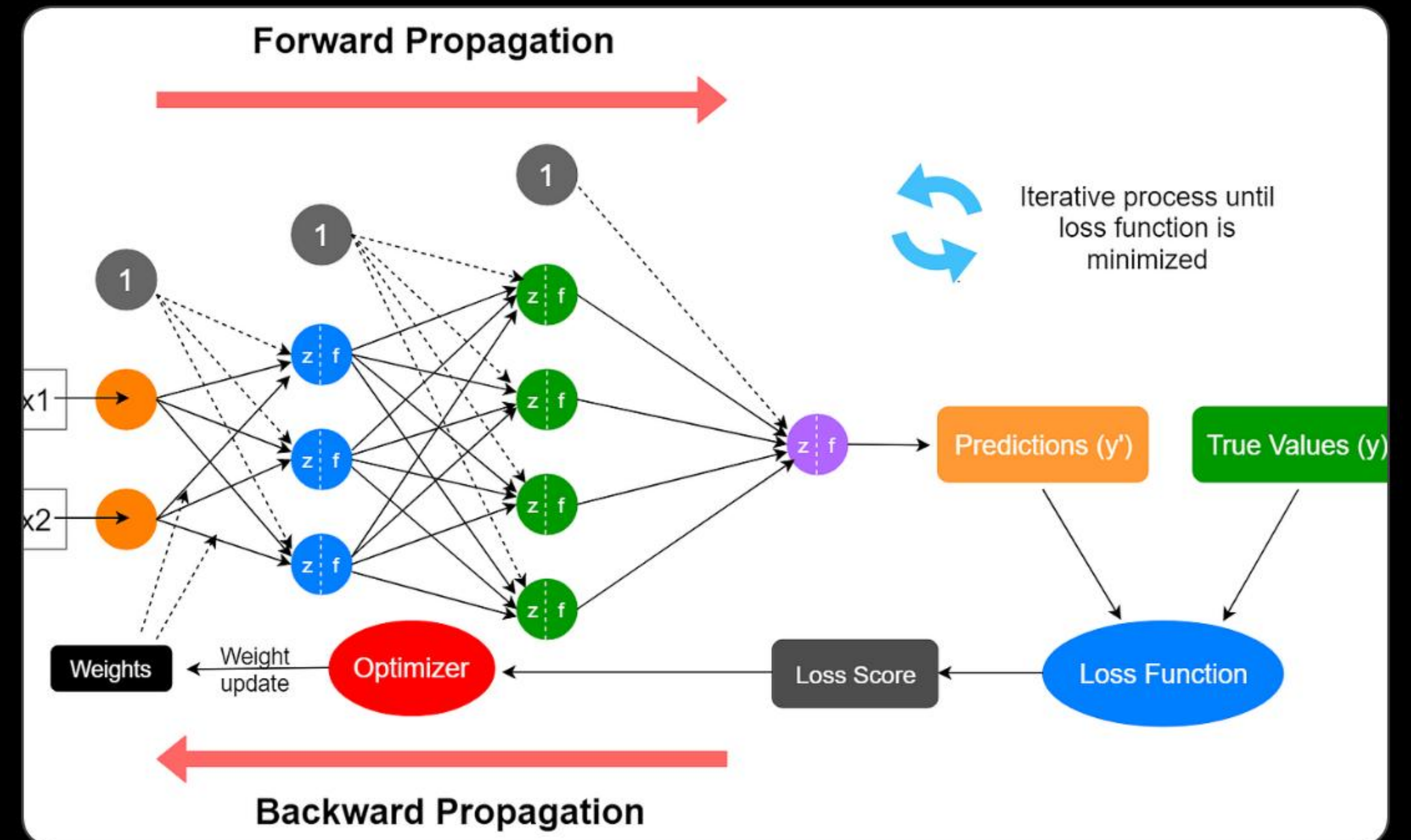
An end-to-end modular architecture designed for high-performance data processing and real-time inference.

# | 1. DATA PREPROCESSING PIPELINE

## AUTOMATED ENGINEERING

The `data_preprocessing.py` module implements a robust cleaning sequence to prepare raw survey data for modeling.

- **Feature Mapping:** Strategic encoding of 'Gender', 'Customer Type', and 'Class' categories.
- **Handling Discrepancies:** Systematic imputation of missing flight distance and delay values.
- **Normalization:** Scaling numerical attributes to ensure feature parity for the Decision Tree algorithm.



## | 2. MODEL TRAINING & ORCHESTRATION



### PREDICTIVE LOGIC

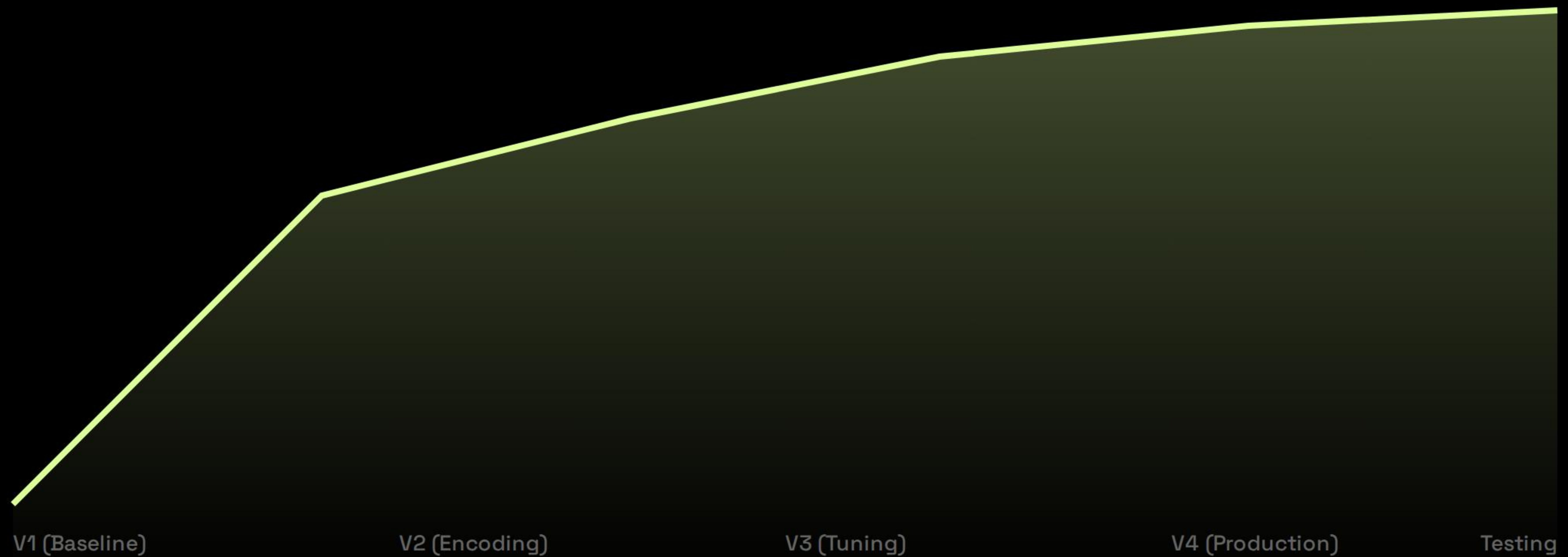
Built on a high-performance **Decision Tree Classifier**, the model identifies non-linear relationships between passenger ratings and overall satisfaction.



### MODULAR STRUCTURE

`main.py` acts as the central orchestrator, seamlessly linking preprocessing, training, and evaluation modules in a production-ready workflow.

# | MODEL PERFORMANCE BENCHMARKS



*Progressive accuracy improvement from baseline (72%) to production-ready (93%).*

## | 3. BUSINESS INTELLIGENCE ENGINE



### **AUTOMATED REPORTS**

The `insights.py` module automatically compiles technical metrics into a structured PDF report for stakeholders.



### **CHARACTER SAFETY**

Proprietary `safe()` encoding routine prevents PDF crashes by handling non-standard UTF-8 characters via Latin-1 fallbacks.



### **CATEGORY MAPPING**

Dynamic re-mapping of encoded IDs back to human-readable labels (e.g., Business, Eco Plus) for transparent reporting.



## 4. REAL-TIME PREDICTION DASHBOARD

### INTERACTIVE UX

The `app.py` dashboard, built with Streamlit, allows stakeholders to simulate passenger experiences through a "What-If" prediction interface.

**Optimized Performance:** Utilizes `@st.cache_resource` to maintain lightning-fast response times during high-frequency inference sessions.

# | CRITICAL SATISFACTION DRIVERS

| Service Feature        | Correlation | Impact Level | Strategic Action                |
|------------------------|-------------|--------------|---------------------------------|
| Online Boarding        | 0.52        | Critical     | Streamline mobile app check-in  |
| Inflight Entertainment | 0.48        | High         | Upgrade Economy cabin hardware  |
| Seat Comfort           | 0.45        | High         | Refurbish older fleet interiors |
| Inflight Wifi          | 0.38        | Medium       | Standardize broadband speeds    |

## | 5. PRODUCTION VALIDATION

93%

TRAINING ACCURACY

### KEY SUCCESS INDICATORS

- ✓ Zero memory leaks via `plt.close()` integration.
- ✓ Balanced performance across all Travel Classes.
- ✓ Robust cross-validation of survey parameters.

# | STRATEGIC ROADMAP

- ↗ **Incentivize Digital Adoption:** Higher satisfaction in "Online Boarding" users suggests a direct link between tech-adoption and positive experience.
- 🪑 **Economy Seating Focus:** Data shows Business class remains highly satisfied; target "Eco Plus" seat comfort to reduce customer churn.
- ⚡ **Real-time Interventions:** Integrate model predictions into ground staff tablets to proactively assist potentially dissatisfied flyers.

# QUESTIONS?

Deep Dive into the ML Pipeline Architecture

**Lead Developer:** Adangatha Noob

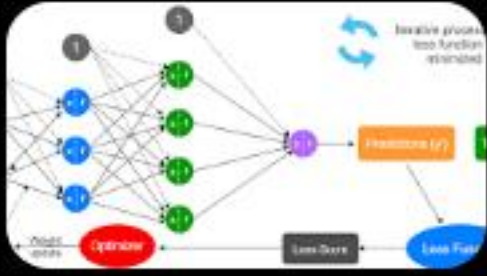
**Project:** Airline Passenger Satisfaction Dashboard



# THANK YOU

Safe Travels and Data-Driven Horizons

# | IMAGE SOURCES



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